# Shortcut learning from image artifacts, region-of-interest masking and its alternatives in medical image classification: a review of the 2021–2026 literature

Literature review for the MSc thesis *Where the Shortcut Sits*

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**Abstract**

This review collects the peer-reviewed work published from January 2021 to October 2026 that bears on one question: when an image artifact that is correlated with the diagnosis lies inside or outside the region of interest (ROI), does restricting the classifier to the ROI remove the shortcut, and what should be used instead? We included 69 studies from journals and conferences of recognised standing (Nature, Lancet and Science families, Medical Image Analysis, MICCAI, CVPR, ECCV, NeurIPS, ICML, ICLR and others), verified each against its publisher or proceedings page and excluded preprints and workshop papers. The literature establishes firmly that medical classifiers exploit acquisition signals, devices, markings and demographic correlates; it documents artifacts per modality, mostly in dermoscopy and chest radiography; it shows that image backgrounds carry predictive signal and that ROI restriction sometimes helps; and it offers group-robustness, invariance, erasure, explanation-guided and generative mitigations, none of which is uniformly better than standard training. Every masking and background-removal study we found assumes that the shortcut lies outside the ROI. One synthetic-bias study varies the bias's distance from the disease region; none varies an artifact's location relative to the ROI used for masking. The clearest report that removing one shortcut can amplify another comes from natural images. The review ends with a gap analysis that positions the thesis against the closest prior work and states what is and is not new.

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# 1 Purpose and review questions

Deep classifiers for medical images often rely on features that predict the label in the training data without being evidence of disease—*shortcuts* (Geirhos et al., 2020)[†]. A common defence is to segment the lesion or organ and classify only the region of interest. The thesis asks when that defence works, and finds that it depends on where the artifact sits: masking removes an artifact outside the ROI but keeps, and makes the model rely more on, an artifact inside it. This review sets out what was known before and alongside that work. It is organised by six questions:

**RQ1** What evidence is there that medical image classifiers use content that is not pathology (artifacts, devices, acquisition signals, demographic correlates)? (Sec. 3)

**RQ2** Which artifacts are documented in each modality, and where do they lie relative to the ROI? (Sec. 4)

**RQ3** What is known about restricting a model to the ROI by masking, segmentation or background removal? (Sec. 5)

**RQ4** How are shortcuts detected and audited? (Sec. 6)

**RQ5** Which mitigation methods exist, and what does each require? (Sec. 7)

**RQ6** What theory explains which feature a model uses, and how should claims of this kind be evaluated? (Secs. 8, 9)

Section 10 draws the gap analysis and Section 11 lists every included study.

# 2 Method

**Search** Searches were run in September and October 2026 with web and scholarly search engines, PubMed, the publishers' sites and the proceedings of MICCAI (Springer LNCS), CVF open access (CVPR, ICCV), ECVA (ECCV), PMLR (ICML, AISTATS, MIDL), NeurIPS and OpenReview

(ICLR). Query terms combined *shortcut learning, spurious correlation, Clever Hans, confounding, dataset bias* with *medical imaging, dermoscopy, skin lesion, ultrasound, calipers, chest radiograph, chest drain, chest tube, capsule endoscopy* and with *masking, segmentation, background, region of interest, counterfactual, invariance, concept erasure, group robustness*. We then followed references backwards and citations forwards from four anchor papers (DeGrave et al., 2021; Brown et al., 2023; Jiménez-Sánchez et al., 2023; Sourget et al., 2025).

**Inclusion** A study was included if it (i) was published, in final peer-reviewed form, between January 2021 and October 2026; (ii) appeared in a journal or main-track conference of recognised standing; and (iii) addressed one of RQ1–RQ6. For reading convenience we mark venues as **A** (Nature, Lancet and Science families, Medical Image Analysis, Radiology: Artificial Intelligence, JAMIA, Information Fusion; NeurIPS, ICML, ICLR, CVPR, ECCV, AAAI, AISTATS and MICCAI) or **B** (other established peer-reviewed journals and conferences, e.g. European Radiology, European Journal of Cancer, JMIR, Computers in Biology and Medicine, Machine Learning, Patterns, MIDL, ISBI, MLSys). The tiers are an aid for the reader, not a formal ranking.

**Exclusion** Preprints without a peer-reviewed version, workshop papers and papers in journals without a consistent standing in the field were excluded even when relevant. They are listed, uncited, in Appendix A so that the reader can see what was left out and why. Studies of segmentation accuracy alone, and fairness studies without a bearing on the mechanism of shortcut use, were excluded. A small number of works published before 2021 are cited as background; they are marked with † and are not counted.

**Verification** For every included study the title, authors, venue, volume, pages or article number and DOI were checked against the publisher's or proceedings' own page. Each summary below states only what the paper's abstract or main results report; where we draw an inference of our own, the text says so.

**Limitations of the method** This is a structured narrative review by one reviewer, not a PRISMA systematic review: screening counts were not recorded, only English-language work was searched, and coverage of 2026 is necessarily incomplete because indexing lags publication. Statements of the form "we found no study" mean no study within this search.

# 3 Evidence that medical classifiers learn shortcuts (RQ1)

## 3.1 Confounded acquisition and dataset assembly

The early warning came from pneumonia detection, where models identified the hospital of origin from radiographs (Zech et al., 2018)†, and from *hidden stratification*, where overall metrics conceal failures on clinically important subsets (Oakden-Rayner et al., 2020)†. The COVID-19 literature then made the problem unmistakable. DeGrave et al. (2021) showed with saliency maps and generative counterfactuals that radiographic COVID-19 detectors relied on laterality markers, patient positioning and other confounds of the source datasets rather than on lung pathology, and lost much of their performance on external data. A systematic review of 62 COVID-19 imaging models found none suitable for clinical use, citing bias in small datasets and the variability of large, internationally sourced ones (Roberts et al., 2021). Maguolo and Nanni (2021) classified COVID-19 radiographs whose lung fields had been blacked out, showing that the datasets carried label information outside the anatomy the task is about.

### 3.2 Demographic attributes as encoded shortcuts

Deep models predict self-reported race from radiographs, even from degraded images (Gichoya et al., 2022), and chest-radiograph classifiers underdiagnose under-served groups, most strongly at intersections such as Hispanic female patients (Seyyed-Kalantari et al., 2021). Disease models encode protected characteristics in their features (Glocker et al., 2023a), and a chest-radiograph foundation model showed sex- and race-related biases (Glocker et al., 2023b); expert-level vision–language foundation models underdiagnose marginalised groups more than radiologists do (Yang et al., 2025). Yang et al. (2024) found that models which encode demographics more strongly show larger fairness gaps, and that debiasing which is optimal in distribution does not transfer to new sites. Two methods test reliance rather than encoding: *shortcut testing* varies how strongly a sensitive attribute is encoded and measures the effect on fairness (Brown et al., 2023), and Ong Ly et al. (2024) estimate external performance without external data from a model's dependence on shortcuts. Jones et al. (2024) separate three causal mechanisms of dataset bias (prevalence, presentation and annotation) that look alike in the data but call for different mitigations, and Banerjee et al. (2023) review causes, evaluation and mitigation of shortcuts for radiologists. For foundation-model features, pooling data across institutions can hide shortcut-like behaviour that per-site evaluation reveals (Germani et al., 2026).

### 3.3 Controlled, synthetic bias

Several studies inject a known bias to study its effect. Stanley et al. (2024) generate synthetic brain images with known disease and bias effects, place the bias either next to the disease region or far from it (opposite hemisphere), and show that this distance changes the performance gaps and how well three mitigations work (reweighing worked best); a follow-up locates where in a convolutional network the bias is encoded and finds that strong encoding does not always mean the model uses it (Stanley et al., 2025). Sun et al. (2023) contaminate chest radiographs with a tag, hyperintensities or an obstruction and find that explanation methods differ strongly in whether they reveal the spurious signal. Boland et al. (2024) study shortcuts such as text labels added to images and locate where in the network the shortcut is used.

**For the thesis.** The traps of the thesis belong to this controlled tradition. Stanley et al. (2024) are the closest precedent: they vary where a synthetic bias lies relative to the disease region (near or far) and find that it matters. They do not study ROI masking, and their bias is synthetic. The thesis varies an artifact's position relative to the ROI that masking keeps, and measures how masking's benefit changes with it, with pasted and real artifacts.

## 4 Artifact shortcuts by modality (RQ2)

### 4.1 Dermoscopy

Dermoscopy has the richest artifact literature. Ruler scale bars changed the melanoma predictions of a market-approved classifier (Winkler et al., 2021), as surgical skin markings had earlier (Winkler et al., 2019)[†]. Explanation-based audits found colour calibration patches, rulers and other artifacts used by ISIC models (Anders et al., 2022; Pahde et al., 2023), and the approach was later extended to semi-automated bias annotation across four medical tasks (Pahde et al., 2025). Hair occluding the lesion degrades classifiers (Wang et al., 2024), and adding synthetic dark-corner artifacts during training helps more than removing real ones (Pewton et al., 2024). The ISIC collections contain duplicate images across releases, which can leak between training and test sets (Cassidy et al., 2022); the ISIC 2020 set was released with patient-level metadata (Rotemberg et al., 2021). Performance falls on darker skin and uncommon diseases (Daneshjou et al., 2022). Lesion segmentation, the source of dermoscopy ROIs, is reviewed by Mirikharaji et al. (2023).

*Location.* Calibration patches and dark corners lie mostly outside the lesion; rulers usually lie near the image edge and markings around the lesion, sometimes overlapping its border; hair crosses it. The studies we reviewed report the effect of an artifact's presence, not of where it lies.

Table 1: Artifacts documented as shortcuts (2021–2026), with their typical position relative to the ROI used for masking. "Position" is our reading of the artifact type, not a quantity reported by the cited studies.

| Modality | Artifact | Typical position | Studies |
|---|---|---|---|
| Dermoscopy | ruler, scale bar, skin marking | mostly outside; markings can overlap the border | Winkler et al. (2021); Anders et al. (2022); Pahde et al. (2023) |
| | hair | crosses the lesion | Wang et al. (2024) |
| | dark corner, colour patch | outside | Pewton et al. (2024); Anders et al. (2022) |
| Ultrasound | calipers, on-screen text | on or around the ROI | Lin et al. (2024) (segmentation only) |
| Chest radiograph | chest tube or drain | inside the lung field | Rueckel et al. (2021); Jiménez-Sánchez et al. (2023) |
| | laterality marker, text, positioning | outside | DeGrave et al. (2021) |
| | source-dataset signal | everywhere, incl. outside | Maguolo and Nanni (2021); Sourget et al. (2025) |
| Capsule endoscopy | bubbles, debris | can overlap the lesion | none found |

### 4.2 Ultrasound

In fetal ultrasound, calipers and on-screen text act as shortcuts even for segmentation (Lin et al., 2024). The public thyroid datasets used in the thesis (Gong et al., 2021; Hou et al., 2024) contain sonographer calipers, mostly on or around the nodule (our observation; the dataset papers do not study them). The closest work we found is a thyroid validation study that considered calipers as a possible confound and judged that differences in their appearance between scanners did not explain its performance difference (Weng et al., 2023). Within our search, no peer-reviewed study measures calipers as a *classification* shortcut in thyroid or ovarian ultrasound, or relates the caliper's position to the nodule ROI.

### 4.3 Chest radiography

Pneumothorax classifiers detect the chest tube that treats the condition. Rueckel et al. (2021) reduced this confounding by training with in-image annotations, and Jiménez-Sánchez et al. (2023) labelled drains to expose the shortcut on a public dataset. A meta-analysis of deep pneumothorax detection reports performance comparable to physicians but a high risk of bias in most studies (Sugibayashi et al., 2023). Saliency maps localise chest pathology worse than radiologists (Saporta et al., 2022), which limits explanation-based audits. Diffusion-based editing allows stress tests for acquisition, manifestation and population shift without new data (Pérez-García et al., 2024). Bassi et al. (2024) penalise relevance outside segmentation masks to make chest radiograph classifiers robust to background bias.

*Location.* A drain projects over the lung field, so for lung-field masks it is an in-ROI artifact. Lung masking therefore cannot remove it: this is the boundary case discussed in the thesis.

### 4.4 Capsule endoscopy and other modalities

Pahde et al. (2025) include a gastrointestinal abnormality task among their four. We found no peer-reviewed study of bubbles or debris as a classification shortcut in capsule endoscopy, a modality in which the artifact and the lesion can share the same pixels.

# 5 Restricting the model to the region of interest (RQ3)

## 5.1 Backgrounds carry signal

In natural images, ImageNet classifiers recognise objects from backgrounds alone, and more accurate models rely less on the background (Xiao et al., 2021); foreground and background sensitivity varies strongly with training and architecture (Moayeri et al., 2022). The same holds in medicine. COVID-19 radiographs remain classifiable with the lungs removed (Maguolo and Nanni, 2021), and in a systematic masking study on chest radiographs and fundus images, models trained on full images performed better on images *without* the ROI than on images that contained only the ROI (Sourget et al., 2025).

## 5.2 ROI restriction as a remedy

Training on segmented dermoscopy images improved balanced accuracy for classifiers trained on HAM10000 but not for those trained on ISIC (Maron et al., 2021). ISNet uses segmentation masks only at training time, penalising relevance outside them, and improves robustness to background bias on chest radiographs (Bassi et al., 2024); in-image annotations serve a similar purpose for chest tubes (Rueckel et al., 2021). Each of these remedies assumes that the shortcut lies outside the region that is kept.

## 5.3 Removing one shortcut can amplify another

Li et al. (2023) showed in natural images that when several shortcuts are present, mitigating one makes the model rely more on the others, and proposed a last-layer ensemble against it. In our reading, ROI masking is a mitigation of exactly this kind: it removes background shortcuts and, if an artifact remains inside the ROI, the same dilemma predicts greater reliance on it. Li et al. did not study masking or medical images.

**For the thesis.** The prior work supports both halves of the problem—backgrounds carry shortcuts, and masking often helps—but every ROI-restriction study assumes the shortcut is in the background. Within our search, no study varies the artifact's location relative to the ROI used for masking (the closest, Stanley et al., 2024, varies a synthetic bias's distance from the disease region without masking), and none tests whether masking increases reliance on an in-ROI artifact in medical images.

# 6 Detecting and auditing shortcuts (RQ4)

**Explanations** Spectral relevance analysis clusters explanation heatmaps to reveal Clever-Hans strategies in ImageNet and ISIC 2019 (Anders et al., 2022); Reveal-to-Revise turns this into an iterative cycle of revealing, modelling, removing and re-evaluating a bias (Pahde et al., 2023, 2025). Explanations are not reliably faithful, however: detection of injected confounders differs widely between methods (Sun et al., 2023), saliency localises chest pathology worse than radiologists (Saporta et al., 2022), and shortcut dependence can be traced to particular layers (Boland et al., 2024). Our inference: a heatmap cannot separate an in-ROI artifact from the pathology it overlaps, because both occupy the same pixels.

**Counterfactuals and editing** Diffusion-based counterfactuals add or remove a suspected shortcut and measure the change in prediction, with a masking scheme that confines the edit to a small region (Weng et al., 2024); RadEdit edits chest radiographs under multiple masks to stress-test classifiers (Pérez-García et al., 2024); and DeGrave et al. (2021) used generative counterfactuals to confirm the COVID-19 confounds.

**Statistical tests** Shortcut testing (Brown et al., 2023), estimation of external performance from internal data (Ong Ly et al., 2024), artifact labels on public datasets (Jiménez-Sánchez et al., 2023) and per-subgroup evaluation (Yang et al., 2024) probe reliance without heatmaps.

**For the thesis.** The thesis uses the counterfactual idea in its simplest form, pasting a real artifact into the same image inside or outside the ROI, which yields a same-image test of location rather than of presence.

# 7 Mitigation (RQ5)

## 7.1 Group robustness

With group labels (artifact × class), GroupDRO (Sagawa et al., 2020)†, variance-of-risk penalties (V-REx; Krueger et al., 2021) and last-layer retraining on a group-balanced set (DFR; Kirichenko et al., 2023) are standard. Izmailov et al. (2022) showed that ERM features already contain much of the core information, so retraining only the last layer can recover robustness, which motivates linear heads on frozen features. Without group labels, JTT up-weights the errors of a first model (Liu et al., 2021) and AFR reweights the last layer automatically (Qiu et al., 2023). Large benchmarks temper expectations: across 20 algorithms and 12 datasets, methods improved subgroup robustness only for some types of shift (Yang et al., 2023), and in medical imaging, bias-mitigation algorithms did not significantly improve fairness over ERM, while model selection mattered more (Zong et al., 2023).

## 7.2 Auxiliary labels and invariance

Counterfactual invariance formalises what a shortcut-free predictor should satisfy; in the anti-causal setting, in which the label causes the image, it implies that the prediction is independent of the nuisance *given the label* (Veitch et al., 2021). Makar et al. (2022) enforce the conditional independences implied by the causal graph with auxiliary labels available at training time, and NuRD trains under a distribution in which the nuisance–label relation is broken (Puli et al., 2022). Kaur et al. (2023) show that the correct independence constraint depends on the data-generating process and that no single constraint suits all shifts, consistent with the mechanism-specific view of Jones et al. (2024).

## 7.3 Concept erasure

Linear erasure removes a concept from the features: iterative nullspace projection (Ravfogel et al., 2020)†, adversarial erasure (Ravfogel et al., 2022) and the closed-form LEACE (Belrose et al., 2023). When the concept is correlated with the label, erasure also removes label signal; SPLINCE preserves covariance with the task label while removing the concept (Holstege et al., 2025).

## 7.4 Explanation-guided and data-centric methods

Class-artifact compensation (Anders et al., 2022), gradient penalties along a concept direction in latent space (Dreyer et al., 2024) and relevance penalties outside a mask (Bassi et al., 2024) use explanations to suppress a known bias. On the data side, in-image annotations (Rueckel et al., 2021), artifact augmentation (Pewton et al., 2024), diffusion-generated synthetic data, which improved fairness out of distribution in histopathology, chest radiography and dermatology (Ktena et al., 2024), and counterfactual removal of shortcuts (Weng et al., 2024) change what the model sees.

**For the thesis.** The thesis's recommended remedy, a class-conditional mean constraint on the masked features, is the first-moment, linear case of the conditional independence of Veitch et al. (2021) and Makar et al. (2022). What the thesis adds is its derivation from the location law, its use after masking, and a registered test of whether any remedy *dominates* masking; the benchmark results of Yang et al. (2023) and Zong et al. (2023) make it unsurprising that none does.

Table 2: Mitigation families and what they need. "Thesis arm" names the method of the thesis that belongs to each family (ERM: standard training; mask: ROI masking).

| Family | Representative studies | Needs at training | Evidence in medical imaging | Thesis arm |
|---|---|---|---|---|
| ROI restriction | Maron et al. (2021); Bassi et al. (2024) | ROI masks | mixed; assumes background shortcut | mask |
| Group reweighting / DRO | Krueger et al. (2021); Kirichenko et al. (2023); Qiu et al. (2023) | group labels (AFR, JTT: none) | no consistent gain over ERM (Zong et al., 2023) | group balancing |
| Conditional invariance | Veitch et al. (2021); Makar et al. (2022); Puli et al. (2022) | auxiliary (artifact) labels | limited | class-conditional constraint |
| Concept erasure | Belrose et al. (2023); Holstege et al. (2025) | concept labels or pairs | limited | U-MtE (label-free) |
| Explanation-guided | Anders et al. (2022); Dreyer et al. (2024); Pahde et al. (2023) | concept annotation | dermoscopy, chest radiograph | — |
| Generative / data | Ktena et al. (2024); Weng et al. (2024) | a generative model | histopathology, chest radiograph, dermoscopy | — |

# 8 Theory of feature choice (RQ6, part 1)

Why does a model pick a shortcut when a stable feature is available? Nagarajan et al. (2021) show that even when the invariant feature fully predicts the label, geometric and statistical skews push ERM toward spurious ones. Gradient descent with cross-entropy can starve less salient features (Pezeshki et al., 2021), and the max-margin bias of default ERM produces shortcut reliance even when the shortcut adds no information (Puli et al., 2023). Hermann et al. (2024) separate a feature's *predictivity* from its *availability* (how easily it is extracted) and quantify shortcut bias as reliance on the more available but less predictive feature. Invariant risk minimisation can fail outside the linear regime or with few environments (Rosenfeld et al., 2021). For frozen features, linear probing can generalise better out of distribution than fine-tuning, which distorts good pretrained features (Kumar et al., 2022).

**For the thesis.** The thesis's linear-Gaussian account fits this line of work. Masking removes competing evidence outside the ROI, which in Hermann et al.'s terms changes the relative predictivity of what remains, so reliance on an in-ROI artifact grows. Unlike the explanatory theories above, the account was used to register quantitative predictions before the results existed.

# 9 Evaluation methodology (RQ6, part 2)

Methodological failures in medical-imaging machine learning, from dataset bias to publication incentives, are reviewed by Varoquaux and Cheplygina (2022), and data leakage, including non-independent splits, inflated results in at least 294 papers across 17 fields (Kapoor and Narayanan, 2023). Variance from data sampling, initialisation and hyperparameters changes benchmark conclusions, and accounting for several sources at once estimates it better (Bouthillier et al., 2021). In MICCAI papers, confidence intervals are rarely reported and their typical width exceeds the gap between the first- and second-ranked methods (Christodoulou et al., 2024). Metrics should follow from the clinical question (Maier-Hein et al., 2024), and reporting should follow CLAIM (Tejani et al., 2024). Model selection can matter more than the mitigation algorithm (Zong et al., 2023; Yang et al., 2023), and subgroup failures are hidden by overall metrics (Yang et al., 2024), as hidden stratification had shown (Oakden-Rayner et al., 2020)[†].

**For the thesis.** These studies motivate the evaluation choices of the thesis: patient- or lesion-grouped splits; a bootstrap that resamples training seeds and test images jointly, since a test set shared by several seeds is one sample, not several; Holm-corrected families; registration before results; and evaluation on shortcut-conflicting cases as well as on the whole unaltered test set.

## 10 Gap analysis: where the thesis stands

Table 3 sets each contribution of the thesis against the closest peer-reviewed work found.

Table 3: Gap analysis. "Open" means not addressed by any included study.

| Question | Established in 2021–2026 | Open | Thesis contribution |
|---|---|---|---|
| Where does the shortcut sit? | Artifacts are documented by presence (Winkler et al., 2021; Rueckel et al., 2021; Lin et al., 2024; Wang et al., 2024); backgrounds carry signal (Maguolo and Nanni, 2021; Sourget et al., 2025; Xiao et al., 2021); a synthetic bias's distance from the disease region changes performance gaps (Stanley et al., 2024) | Location relative to the masking ROI as an experimental variable, with real artifacts | Overlap $r$ controlled in sweeps; real traps with in-ROI and out-of-ROI artifacts; transplant of the same real artifact across the ROI boundary of the same images |
| Does ROI masking fix it? | Sometimes (Maron et al., 2021; Bassi et al., 2024); every study assumes a background shortcut | When and why masking fails | Location law: masking removes out-of-ROI shortcuts and leaves in-ROI ones, in four cohorts and three modalities |
| Can a mitigation amplify a shortcut? | Yes, for multiple shortcuts in natural images (Li et al., 2023) | Whether masking amplifies an in-ROI artifact in medical images, and by what mechanism | Reliance on the in-ROI artifact grows after masking because competing evidence is removed |
| Can it be predicted? | Explanatory theories of feature choice (Nagarajan et al., 2021; Puli et al., 2023; Hermann et al., 2024) | Quantitative, prospectively tested predictions | Two-parameter linear-Gaussian account; predictions registered before the results |
| Is harm visible clinically? | Overall metrics hide subgroup failures (Yang et al., 2024); see also Oakden-Rayner et al. (2020)† | Effect of masking on shortcut-conflicting cases of unaltered data | Lower AUROC on conflicting cases of patient-disjoint and new-patient thyroid data; sensitivity loss among malignant nodules with calipers |

| Question | Established in 2021–2026 | Open | Thesis contribution |
|---|---|---|---|
| What to use instead? | No method uniformly beats ERM (Yang et al., 2023; Zong et al., 2023); conditional invariance with auxiliary labels (Veitch et al., 2021; Makar et al., 2022); erasure (Belrose et al., 2023; Holstege et al., 2025) | Remedies tested against masking under a registered criterion | 42-component dominance criterion, two registered rounds; no remedy dominates; the class-conditional constraint comes closest |
| Which modalities? | Mostly dermoscopy and chest radiography; calipers in fetal-US segmentation (Lin et al., 2024) | Thyroid and ovarian ultrasound calipers, capsule debris as classification shortcuts | Traps in thyroid and ovarian ultrasound and capsule endoscopy, alongside dermoscopy |
| How strong is the evidence? | CIs rarely reported (Christodoulou et al., 2024); seed and data variance (Bouthillier et al., 2021); leakage (Kapoor and Narayanan, 2023) | Joint seed × image uncertainty for shortcut experiments | Crossed bootstrap, Holm families, registration, every interval traced to a result file |

**What is not new** That a mask cannot remove what it keeps is self-evident; what was not established is that masking makes reliance on a kept artifact *grow*, how large this is on real artifacts, that it is caused by location, and how to predict it. Group balancing, last-layer retraining, IRM and V-REx are established methods used as baselines. The class-conditional constraint is a special case of known conditional-invariance regularisers (Veitch et al., 2021; Makar et al., 2022), and disease-preserving erasure overlaps with SPLINCE (Holstege et al., 2025); the thesis presents both as adaptations, not as new methods.

**Threats to this positioning** (i) Two workshop papers study masking directly: early versus late background masking for fine-grained recognition, and artifact-based domain generalisation with test-time masking in dermoscopy (Appendix A). Both assume background shortcuts, but a reviewer may know them. (ii) The whac-a-mole dilemma (Li et al., 2023) anticipates the amplification conceptually; the thesis should cite it and state that it supplies the location-specific mechanism and the medical evidence. (iii) Coverage of 2026 is incomplete; the review should be repeated before submission.

## 11 Summary of included studies

Table 4: All 69 included studies (2021–2026), grouped by theme. Tier: A or B as defined in Sec. 2. CXR: chest radiograph; US: ultrasound; NI: natural images; –: not modality-specific.

| Study | Venue | Tier | Modality | Finding or contribution | Relevance to the thesis |
|---|---|---|---|---|---|
| *Evidence of shortcut learning and bias* | | | | | |
| DeGrave et al. (2021) | Nat. Mach. Intell. | A | CXR | COVID-19 detectors use dataset confounds | motivating example |
| Roberts et al. (2021) | Nat. Mach. Intell. | A | CXR, CT | none of 62 models clinically usable | dataset bias |
| Maguolo and Nanni (2021) | Inf. Fusion | A | CXR | classification with lungs blacked out | signal outside the ROI |
| Seyyed-Kalantari et al. (2021) | Nat. Med. | A | CXR | underdiagnosis of under-served groups | hidden subgroup harm |

| Study | Venue | Tier | Modality | Finding or contribution | Relevance to the thesis |
|---|---|---|---|---|---|
| Gichoya et al. (2022) | Lancet Digit. Health | A | multiple | race predictable from images | invisible correlates |
| Glocker et al. (2023a) | eBioMedicine | A | CXR | disease models encode protected attributes | encoding vs use |
| Glocker et al. (2023b) | Radiol. AI | A | CXR | bias in a foundation model | frozen-feature heads |
| Brown et al. (2023) | Nat. Commun. | A | derm., CXR | shortcut testing | test of reliance |
| Banerjee et al. (2023) | J. Am. Coll. Radiol. | B | radiology | review of causes and mitigation | background |
| Yang et al. (2024) | Nat. Med. | A | multiple | fairness does not transfer to new sites | evaluation on new data |
| Yang et al. (2025) | Sci. Adv. | A | CXR | VLM foundation models underdiagnose | foundation models |
| Jones et al. (2024) | Nat. Mach. Intell. | A | – | three causal bias mechanisms | choice of remedy |
| Ong Ly et al. (2024) | npj Digit. Med. | A | medical | generalisation estimated without external data | reliance estimation |
| Stanley et al. (2024) | JAMIA | A | brain MRI | synthetic bias near or far from the disease region; distance matters; reweighing best | closest location precedent |
| Stanley et al. (2025) | eBioMedicine | A | brain MRI | where bias is encoded | encoding vs use |
| Sun et al. (2023) | MICCAI | A | CXR | explanations vary in detecting injected confounders | limits of heatmaps |
| Boland et al. (2024) | MIDL | B | medical | layer-wise shortcut localisation | detection |
| Sourget et al. (2025) | J. Imaging Inform. Med. | B | CXR, fundus | ROI-removed images remain classifiable | closest masking study |
| Germani et al. (2026) | Med. Image Anal. | A | medical | pooled data hide shortcut-like behaviour | foundation features |
| Ktena et al. (2024) | Nat. Med. | A | histo., CXR, derm. | synthetic data improve OOD fairness | generative remedy |
| *Dermoscopy* | | | | | |
| Winkler et al. (2021) | Eur. J. Cancer | B | derm. | scale bars alter a market-approved CNN | artifact by presence |
| Maron et al. (2021) | JMIR | B | derm. | segmentation helps on HAM, not ISIC | masking evidence (mixed) |
| Daneshjou et al. (2022) | Sci. Adv. | A | derm. (clinical) | lower accuracy on dark skin | distribution shift |
| Cassidy et al. (2022) | Med. Image Anal. | A | derm. | ISIC duplicates and benchmarks | grouped splits |
| Mirikharaji et al. (2023) | Med. Image Anal. | A | derm. | survey of lesion segmentation | source of ROI masks |
| Wang et al. (2024) | Comput. Biol. Med. | B | derm. | hair degrades classifiers | in-ROI artifact |
| Pewton et al. (2024) | Comput. Methods Programs Biomed. | B | derm. | adding dark corners beats removal | out-of-ROI artifact |
| Anders et al. (2022) | Inf. Fusion | A | derm., NI | SpRAy and class-artifact compensation | detection and remedy |
| Pahde et al. (2023) | MICCAI | A | derm., X-ray | Reveal-to-Revise life cycle | detection and remedy |
| Pahde et al. (2025) | Mach. Learn. | B | 4 tasks | semi-automated bias annotation | artifact labels |
| *Ultrasound and chest radiography* | | | | | |
| Lin et al. (2024) | MICCAI | A | US | calipers and text as segmentation shortcuts | ultrasound calipers |

| Study | Venue | Tier | Modality | Finding or contribution | Relevance to the thesis |
|---|---|---|---|---|---|
| Weng et al. (2023) | Clin. Imaging | B | US | thyroid model validated on other scanners; caliper differences judged not to explain results | calipers as a confound |
| Rueckel et al. (2021) | Eur. Radiol. | B | CXR | annotations reduce chest-tube confounding | in-ROI device |
| Sugibayashi et al. (2023) | Eur. Respir. Rev. | B | CXR | meta-analysis; high risk of bias | drain context |
| Jiménez-Sánchez et al. (2023) | IEEE ISBI | B | CXR | drain labels expose a shortcut | chest boundary case |
| Saporta et al. (2022) | Nat. Mach. Intell. | A | CXR | saliency localises worse than radiologists | limits of heatmaps |
| Pérez-García et al. (2024) | ECCV | A | CXR | diffusion editing for stress tests | counterfactual testing |
| Bassi et al. (2024) | Nat. Commun. | A | CXR | relevance penalties outside masks | mask-based remedy |
| *Backgrounds and masking in natural images* | | | | | |
| Xiao et al. (2021) | ICLR | A | NI | backgrounds alone predict the class | background shortcuts |
| Moayeri et al. (2022) | CVPR | A | NI | foreground/background sensitivity | masking evaluation |
| Li et al. (2023) | CVPR | A | NI | mitigating one shortcut amplifies others | closest to amplification |
| *Detection by counterfactuals and explanation-guided training* | | | | | |
| Weng et al. (2024) | ECCV | A | NI, medical | diffusion counterfactuals add/remove shortcuts | same-image test |
| Dreyer et al. (2024) | AAAI | A | derm., NI | latent gradient penalisation | explanation remedy |
| *Mitigation: group robustness and benchmarks* | | | | | |
| Krueger et al. (2021) | ICML | A | – | V-REx | baseline |
| Liu et al. (2021) | ICML | A | – | JTT, no group labels | label-free baseline |
| Izmailov et al. (2022) | NeurIPS | A | – | ERM features hold core information | frozen features |
| Kirichenko et al. (2023) | ICLR | A | – | last-layer retraining (DFR) | balanced head |
| Qiu et al. (2023) | ICML | A | – | automatic feature reweighting | label-free baseline |
| Yang et al. (2023) | ICML | A | mixed | no method best across shifts | no dominance expected |
| Zong et al. (2023) | ICLR | A | multiple | mitigation ≈ ERM in medicine | no dominance expected |
| *Mitigation: invariance and erasure* | | | | | |
| Veitch et al. (2021) | NeurIPS | A | – | counterfactual invariance | basis of the constraint |
| Makar et al. (2022) | AISTATS | A | – | auxiliary-label conditional independence | closest remedy |
| Puli et al. (2022) | ICLR | A | NI, CXR | nuisance-randomised distillation | alternative remedy |
| Kaur et al. (2023) | ICLR | A | – | constraint depends on the causal graph | why class-conditional |
| Ravfogel et al. (2022) | ICML | A | – | adversarial linear erasure | erasure family |
| Belrose et al. (2023) | NeurIPS | A | – | closed-form linear erasure | erasure baseline |
| Holstege et al. (2025) | NeurIPS | A | – | erasure preserving task signal | disease-preserving erasure |
| *Theory* | | | | | |
| Nagarajan et al. (2021) | ICLR | A | – | skews drive spurious reliance | theory |
| Rosenfeld et al. (2021) | ICLR | A | – | limits of IRM | IRM baseline |
| Pezeshki et al. (2021) | NeurIPS | A | – | gradient starvation | theory |

| Study | Venue | Tier | Modality | Finding or contribution | Relevance to the thesis |
|---|---|---|---|---|---|
| Kumar et al. (2022) | ICLR | A | – | linear probing vs fine-tuning OOD | frozen heads |
| Puli et al. (2023) | NeurIPS | A | – | max-margin causes shortcuts | theory |
| Hermann et al. (2024) | ICLR | A | – | availability vs predictivity | closest to the account |
| *Evaluation methodology and reporting* | | | | | |
| Varoquaux and Cheplygina (2022) | npj Digit. Med. | A | – | methodological failures | study design |
| Kapoor and Narayanan (2023) | Patterns | B | – | leakage taxonomy | grouped splits |
| Bouthillier et al. (2021) | MLSys | B | – | sources of benchmark variance | crossed bootstrap |
| Christodoulou et al. (2024) | MICCAI | A | – | CIs rarely reported | intervals everywhere |
| Maier-Hein et al. (2024) | Nat. Methods | A | – | problem-driven metrics | metric choice |
| Tejani et al. (2024) | Radiol. AI | A | – | CLAIM 2024 checklist | reporting |

# A Identified but excluded by the venue criterion

These items are relevant but do not meet the venue criterion of Sec. 2: preprints, workshop papers or journals outside it. They are listed for transparency and are not counted.

- Aniraj, Dantas, Ienco, Marcos (2023), *Masking strategies for background bias removal in computer vision models*, ICCV Workshops (OODCV): early (image-level) masking gave the best out-of-distribution accuracy in fine-grained recognition. The closest general-vision masking study; workshop paper.
- Bissoto, Barata, Valle, Avila (2022), *Artifact-based domain generalization of skin lesion models*, ECCV Workshops (ISIC); and Bissoto et al. (2023), *Test-time selection for robust skin lesion analysis*, MICCAI Workshops (ISIC). Workshop papers.
- Boland et al. (2024), *All you need is a guiding hand: mitigating shortcut bias in deep learning models for medical imaging*, MICCAI FAIMI workshop (LNCS 15198).
- Nauta, Walsh, Dubowski, Seifert (2022), *Uncovering and correcting shortcut learning in machine learning models for skin cancer diagnosis*, *Diagnostics* (MDPI).
- Jin, Gerych, Ghassemi (2024), *MaskMedPaint*, arXiv:2411.10686 (preprint).
- Pedersen, Sydendal, Cheplygina, Sourget (2026), real-world chest X-ray shortcuts in MedCLIP, arXiv:2608.12086 (preprint).
- Kina, Petersen (2026), *Prevalence calibration as shortcut mitigation*, arXiv:2609.07922 (preprint).
- Chuang et al. (2023), *Debiasing vision-language models via biased prompts*, arXiv:2302.00070 (preprint).
- Arjovsky, Bottou, Gulrajani, Lopez-Paz (2019), *Invariant risk minimization*, arXiv:1907.02893: a preprint and before 2021, but the source of the IRMv1 baseline used in the thesis.

# B Datasets and encoders cited for provenance

Cited for the data and models the thesis uses, not as evidence: ISIC 2020 (Rotemberg et al., 2021), TN3K (Gong et al., 2021), the thyroid pathology dataset (Hou et al., 2024), DINOv2 (Oquab et al., 2024), RAD-DINO (Pérez-García et al., 2025) and PanDerm (Yan et al., 2025).

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